

Movement Matters:

How A Virtual Reality Learning System Facilitates US Geographic Learning for International Students

Yupei Duan¹ , Xinhao Xu¹ , Shangman Li¹ , Lanxin Xue¹ , Hao He² , Yuanyuan Gu¹

University of Missouri-Columbia¹, USA. Emporia State University, USA²



Equip Students to Engage With the Interconnected World

- **Fosters ethical decision-making**

(Solem & Weiguo, 2018)

- **Enhances awareness of global issues and cultures**

(Niu et al., 2023)

- **Promotes international peace and understanding**

(Lidstone & Stoltman, 2002)



Virtual Reality (VR) and Embodied Learning

VR

- Immersive and interactive.
- Users can explore virtual spaces as if they were physically present.
- Enhances learning through realistic simulations and hands-on activities.

Embodied Learning

- Connects physical movement with cognitive processes.
- Emphasizes learning by doing.
- Allows students to engage actively
- Makes abstract concepts easier to understand.

(Chatain et al., 2023; Johnson-Glenberg et al., 2020; Xu et al., 2022)



Learning System Design



Figure 1 & 2. A participant uses VirtualGeo to learn and to draw a map.



Research Questions

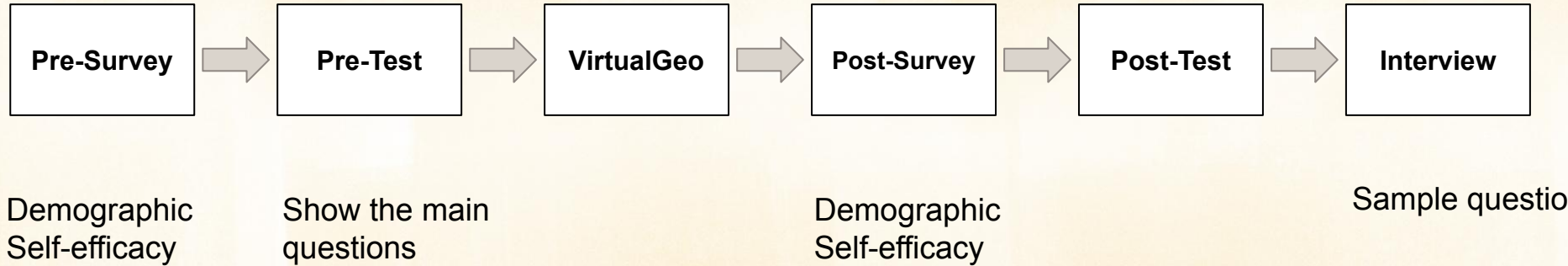
Research Question 1 (RQ1): How does VirtualGeo improve participants' US geographic knowledge by comparing pre- and post-test?

Research Question 2 (RQ2): How do international students interpret their embodied experience and cognitive load while participating in the VirtualGeo intervention?

Research Question 3 (RQ3): How does students' embodied experience correlate with their perceived cognitive load?



Research Design



Sampling

Characteristics	Category	Participants sample	
		n	%
Gender	Male	4	57%
	Female	3	43%
Age	24-28	3	42%
	29-33	2	29%
	34-38	2	8%
Familiarity about U.S. Geography	Not familiar at all	3	43%
	Slightly familiar	3	43%
	Moderately familiar	1	14%
	Very Familiar	0	0%
Experience with Virtual Reality	Yes	3	43%
	No	4	57%

Table 1. Demographic characteristics of the participants (n=7)



Results

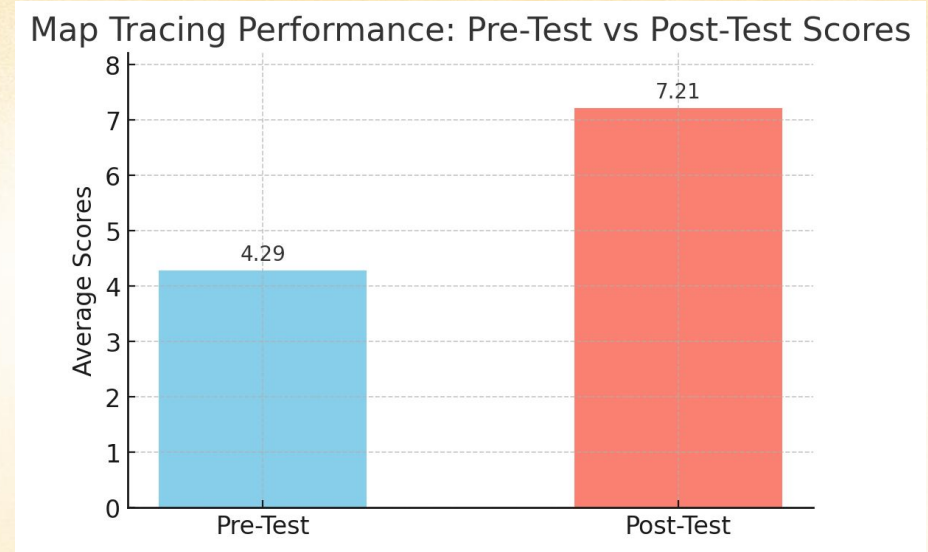
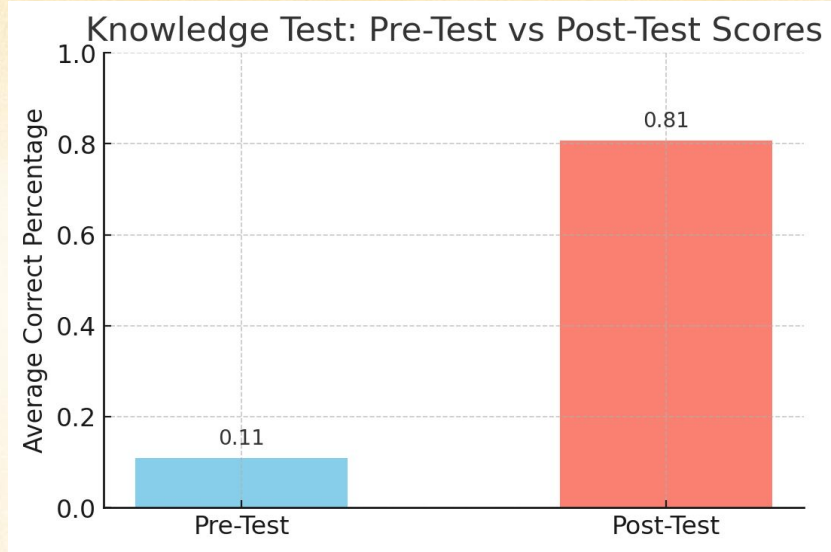


Figure 3 & 4. Improvement of U.S. geographic understanding and map drawing skills between pre- and post-test



Results

Positive Evaluations:

All participants reported positive embodied learning experiences.

Correlation Analysis:

- Pearson Correlation:** 0.697

Indicates a moderately strong positive correlation between embodied experience and cognitive load.

- Spearman Correlation:** 0.717

Confirms the moderately strong positive relationship.



Discussion

Explain correlation data, what does it mean.

Implication

Insights: How to promote learning exp and outcomes



Conclusion

Reemphasize the Research questions and significance.

Look to the future.

Limitations and future work



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Thanks!

yupei.duan@Missouri.edu

